

QUESTIONS

Q1

Constructor and Destructor
Execution Order

10 marks

Q2

Constructor Overloading
Resolution

10 marks

Q3

Operator Overloading for
Complex Numbers

10 marks

Q4

Shallow Copy in Hospital
Management System

10 marks

Q5

Polynomial Class with
Operator Overloading

10 marks

Q6

Static Member Behavior in a
Banking System

10 marks

Q7

Friend Function vs Member
Function

10 marks

Q8

Inheritance and Constructor
Chaining

10 marks

Q9

Virtual Functions and Runtime
Polymorphism

10 marks

Q10

Exception Handling in a File
Processing System

10 marks

Q1

**Constructor and Destructor Execution
Order** 10
marks

Consider the following program:

```
class Demo {  
    int x;  
public:  
    Demo(int a) { x = a; cout << "Constructor " << x <<  
        endl; }  
    ~Demo() { cout << "Destructor " << x << endl; }  
};  
  
int main() {  
    Demo d1(10), d2(20);  
}
```

Tasks:

1. Predict the exact output sequence.
2. Analyze the order of constructor and destructor execution.
3. Explain why destructors are called in reverse order.

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CO3_AT1_Analytical Problem Solving

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10 marks



Q2 Constructor Overloading Resolution 10 marks

```
class Product {  
public:  
    Product() { cout << "Default"; }  
    Product(int x) { cout << "One param"; }  
    Product(int x, int y) { cout << "Two param"; }  
};
```

Tasks:

1. Analyze which constructor is called for different object declarations.
2. Predict the output for Product p1; Product p2(10); Product p3(10,20);
3. Explain how the compiler resolves constructor overloading.

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QUESTIONS

- Q1** ✓

Constructor and Destructor Execution Order

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10 marks
- Q8** ✓

Inheritance and Constructor Chaining

10 marks
- Q9** ✓

Virtual Functions and Runtime Polymorphism

10 marks
- Q10** ✓

Exception Handling in a File Processing System

10 marks

Q3

Operator Overloading for Complex Numbers

10 marks

```
class Complex {
    int r, i;
public:
    Complex(int a, int b) { r = a; i = b; }
    Complex operator+(Complex c) {
        return Complex(r + c.r, i + c.i);
    }
};
```

- Tasks:
1. Analyze how the + operator works for objects.
 2. Predict the result of adding two Complex objects.
 3. Compare operator overloading with function-based addition.

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10 marks

Q4

Shallow Copy in Hospital Management System 10 marks

A hospital management system maintains patient records where each Patient object stores dynamically allocated medical history details (using pointers). When patient records are copied for insurance processing or referral to another hospital, the system uses copy constructors.

Tasks:

1. What problems may occur if the system uses a shallow copy while duplicating patient records?
2. How can this lead to issues such as data corruption or accidental modification of medical history?
3. Explain how a deep copy mechanism ensures safe duplication of patient records.

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Virtual Functions and Runtime

Static Member Behavior in a Banking System

10
marks

```
class Account {  
    static int totalAccounts;  
    int accNo;  
public:  
    Account() { totalAccounts++; accNo =  
        totalAccounts; }  
    static int getTotal() { return totalAccounts; }  
};  
int Account::totalAccounts = 0;  
  
int main() {  
    Account a1, a2, a3;  
    cout << Account::getTotal();  
    {  
        Account a4;  
        cout << Account::getTotal();  
    }  
    cout << Account::getTotal();  
}
```

Tasks:

1. Predict the exact output of the program.
2. Analyze why the static member retains its value even after a4 goes out of scope.
3. Explain why getTotal() can be called without creating an object, and justify when static members are appropriate in real-world systems.

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